

OpenStax Calculus Volume 3

Section 5.2 Exercises: Double Integrals over General Regions

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In the following exercises, specify whether the region is of Type I or Type II.

1. The region D bounded by $y = x^3$, $y = x^3 + 1$, $x = 0$, and $x = 1$ as given in the following figure.
[Figure: A region is bounded by $y = 1 + x^3$, $y = x^3$, $x = 0$, and $x = 1$.]
2. Find the average value of the function $f(x, y) = 3xy$ on the region graphed in the previous exercise.
3. Find the area of the region D given in the previous exercise.
4. The region D bounded by $y = \sin x$, $y = 1 + \sin x$, $x = 0$, and $x = \frac{\pi}{2}$ as given in the following figure.
[Figure: A region is bounded by $y = 1 + \sin x$, $y = \sin x$, $x = 0$, and $x = \frac{\pi}{2}$.]
5. Find the average value of the function $f(x, y) = \cos x$ on the region graphed in the previous exercise.
6. Find the area of the region D given in the previous exercise.
7. The region D bounded by $x = y^2 - 1$ and $x = \sqrt{1 - y^2}$ as given in the following figure.
[Figure: A region is bounded by $x = -1 + y^2$ and $x = \sqrt{1 - y^2}$.]
8. Find the volume of the solid under the graph of the function $f(x, y) = xy + 1$ and above the region in the figure in the previous exercise.
9. The region D bounded by $y = 0$, $x = -10 + y$, and $x = 10 - y$ as given in the following figure.
[Figure: A region is bounded by $x = -10 + y$, $x = 10 - y$, and $y = 0$.]
10. Find the signed volume of the solid under the graph of the function $f(x, y) = x + y$ and above the region in the figure from the previous exercise.
11. The region D bounded by $y = 0$, $x = y - 1$, and $x = \frac{\pi}{2}$ as given in the following figure.
[Figure: A region is bounded by $x = \frac{\pi}{2}$, $y = 0$, and $x = -1 + y$.]
12. The region D bounded by $y = 0$ and $y = x^2 - 1$ as given in the following figure.
[Figure: A region is bounded by $y = 0$ and $y = -1 + x^2$.]
13. Let D be the region bounded by the curve $y = 2 - x^2$ and below the equations $y = x$ and $y = -x$. Explain why D is neither of Type I nor Type II.

14. Let D be the region bounded above by the curve of the equation $y = 4 - x^2$ and below by $y = \cos x$ and the x -axis. Explain why D is neither of Type I nor Type II.

In the following exercises, evaluate the double integral $\iint_D f(x, y) dA$ over the region D .

15. $f(x, y) = 2x + 5y$ and $D = \{(x, y) \mid 0 \leq x \leq 1, x^3 \leq y \leq x^3 + 1\}$.
 16. $f(x, y) = 1$ and $D = \{(x, y) \mid 0 \leq x \leq \frac{\pi}{2}, \sin x \leq y \leq 1 + \sin x\}$.
 17. $f(x, y) = 2$ and $D = \{(x, y) \mid 0 \leq y \leq 1, y - 1 \leq x \leq \arccos y\}$.
 18. $f(x, y) = xy$ and $D = \{(x, y) \mid -1 \leq y \leq 1, y^2 - 1 \leq x \leq \sqrt{1 - y^2}\}$.
 19. $f(x, y) = \sin y$ and D is the triangular region with vertices $(0, 0)$, $(0, 3)$, and $(3, 0)$.
 20. $f(x, y) = -x + 1$ and D is the triangular region with vertices $(0, 0)$, $(0, 2)$, and $(2, 2)$.

Evaluate the iterated integrals.

21. $\int_0^1 \int_{2x}^{3x} (x + y^2) dy dx$.
 22. $\int_0^1 \int_{2\sqrt{x}}^{2\sqrt{x}+1} (xy + 1) dy dx$.
 23. $\int_e^{e^2} \int_{\ln u}^2 (v + \ln u) dv du$.
 24. $\int_1^2 \int_{-u^2-1}^{-u} (8uv) dv du$.
 25. $\int_0^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} (2x + 4x^3) dx dy$.
 26. $\int_0^{1/2} \int_{-\sqrt{1-4y^2}}^{\sqrt{1-4y^2}} 4 dx dy$.

27. Let D be the region in the first quadrant bounded by $y = 1 - x^2$, $y = 4 - x^2$, and the x - and y -axes.

- (a) Show that $\iint_D x dA = \int_0^1 \int_{1-x^2}^{4-x^2} x dy dx + \int_1^2 \int_0^{4-x^2} x dy dx$ by dividing the region D into two regions of Type I.
 (b) Evaluate the integral $\iint_D x dA$.

28. Let D be the region bounded by $y = 1$, $y = x$, $y = \ln x$, and the x -axis.

- (a) Show that $\iint_D y dA = \int_0^1 \int_0^x y dy dx + \int_1^e \int_{\ln x}^1 y dy dx$ by dividing D into two regions of Type I.
 (b) Evaluate the integral $\iint_D y dA$.

29. (a) Show that $\iint_D y^2 dA = \int_{-1}^0 \int_{-x}^{2-x^2} y^2 dy dx + \int_0^1 \int_x^{2-x^2} y^2 dy dx$ by dividing the region D into two regions of Type I, where $D = \{(x, y) \mid y \geq x, y \geq -x, y \leq 2 - x^2\}$.
 (b) Evaluate the integral $\iint_D y^2 dA$.

30. Let D be the region bounded by $y = x^2$, $y = x + 2$, and $y = -x$.

- (a) Show that $\iint_D x \, dA = \int_0^1 \int_{-y}^{\sqrt{y}} x \, dx \, dy + \int_1^4 \int_{y-2}^{\sqrt{y}} x \, dx \, dy$ by dividing the region D into two regions of Type II, where $D = \{(x, y) \mid y \geq x^2, y \geq -x, y \leq x + 2\}$.
- (b) Evaluate the integral $\iint_D x \, dA$.
31. The region D bounded by $x = 0$, $y = x^5 + 1$, and $y = 3 - x^2$ is shown in the following figure. Find the area $A(D)$ of the region D .
[Figure: A region is bounded by $y = 1 + x^5$, $y = 3 - x^2$, and $x = 0$.]
32. The region D bounded by $y = \cos x$, $y = 4 + \cos x$, and $x = \pm \frac{\pi}{3}$ is shown in the following figure. Find the area $A(D)$ of the region D .
[Figure: A region is bounded by $y = \cos x$, $y = 4 + \cos x$, $x = -\frac{\pi}{3}$, and $x = \frac{\pi}{3}$.]
33. Find the area $A(D)$ of the region $D = \{(x, y) \mid y \geq 1 - x^2, y \leq 4 - x^2, y \geq 0, x \geq 0\}$.
34. Let D be the region bounded by $y = 1$, $y = x$, $y = \ln x$, and the x -axis. Find the area $A(D)$ of the region D .
35. Find the average value of the function $f(x, y) = \sin y$ on the triangular region with vertices $(0, 0)$, $(0, 3)$, and $(3, 0)$.
36. Find the average value of the function $f(x, y) = -x + 1$ on the triangular region with vertices $(0, 0)$, $(0, 2)$, and $(2, 2)$.

In the following exercises, change the order of integration and evaluate the integral.

37. $\int_{-1}^{\pi/2} \int_0^{x+1} \sin x \, dy \, dx$.
38. $\int_0^1 \int_{x-1}^{1-x} x \, dy \, dx$.
39. $\int_{-1}^0 \int_{-\sqrt{y+1}}^{\sqrt{y+1}} y^2 \, dx \, dy$.
40. $\int_{-1}^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} y \, dx \, dy$.
41. The region D is shown in the following figure. Evaluate the double integral $\iint_D (x^2 + y) \, dA$ by using the easier order of integration.
[Figure: A region is bounded by $y = -4 + x^2$ and $y = 4 - x^2$.]
42. The region D is given in the following figure. Evaluate the double integral $\iint_D (x^2 - y^2) \, dA$ by using the easier order of integration.
[Figure: A region is bounded by $y^4 = 1 - x$ and $y^4 = 1 + x$.]
43. Find the volume of the solid under the surface $z = 2x + y^2$ and above the region bounded by $y = x^5$ and $y = x$.
44. Find the volume in the first octant of the solid under the plane $z = 3x + y$ and above the region determined by $y = x^7$ and $y = x$.
45. Find the volume of the solid under the plane $z = x - y$ and above the region bounded by $x = \tan y$, $x = -\tan y$, and $x = 1$.
46. Find the volume of the solid under the surface $z = x^3$ and above the plane region bounded by $x = \sin y$, $x = -\sin y$, and $x = 1$ for values of y between $y = -\frac{\pi}{2}$ and $y = \frac{\pi}{2}$.

47. Let g be a positive, increasing, and differentiable function on the interval $[a, b]$. Show that the volume of the solid under the surface $z = g'(x)$ and above the region bounded by $y = 0$, $y = g(x)$, $x = a$, and $x = b$ is given by $\frac{1}{2}(g^2(b) - g^2(a))$.
48. Let g be a positive, increasing, and differentiable function on the interval $[a, b]$, and let k be a positive real number. Show that the volume of the solid under the surface $z = g'(x)$ and above the region bounded by $y = g(x)$, $y = g(x) + k$, $x = a$, and $x = b$ is given by $k(g(b) - g(a))$.
49. Find the volume of the solid situated in the first octant and determined by the planes $z = 2$, $z = 0$, $x + y = 1$, $x = 0$, and $y = 0$.
50. Find the volume of the solid situated in the first octant and bounded by the planes $x + 2y = 1$, $x = 0$, $y = 0$, $z = 4$, and $z = 0$.
51. Find the volume of the solid bounded by the planes $x + y = 1$, $x - y = 1$, $x = 0$, $z = 0$, and $z = 10$.
52. Find the volume of the solid bounded by the planes $x + y = 1$, $x - y = 1$, $x + y = -1$, $x - y = -1$, $z = 1$, and $z = 0$.
53. Let S_1 and S_2 be the solids situated in the first octant under the planes $x + y + z = 1$ and $x + y + 2z = 1$, respectively, and let S be the solid situated between S_1 , S_2 , $x = 0$, and $y = 0$.
- Find the volume of the solid S_1 .
 - Find the volume of the solid S_2 .
 - Find the volume of the solid S by subtracting the volumes of the solids S_1 and S_2 .
54. Let S_1 and S_2 be the solids situated in the first octant under the planes $2x + 2y + z = 2$ and $x + y + z = 1$, respectively, and let S be the solid situated between S_1 , S_2 , $x = 0$, and $y = 0$.
- Find the volume of the solid S_1 .
 - Find the volume of the solid S_2 .
 - Find the volume of the solid S by subtracting the volumes of the solids S_1 and S_2 .
55. Let S_1 and S_2 be the solids situated in the first octant under the plane $x + y + z = 2$ and under the sphere $x^2 + y^2 + z^2 = 4$, respectively. If the volume of the solid S_2 is $\frac{4\pi}{3}$, determine the volume of the solid S situated between S_1 and S_2 by subtracting the volumes of these solids.
56. Consider the plane $x + y + z = 2$ and the cylinder $x^2 + y^2 = 4$ in the first octant.
- Find the volume under the plane.
 - Find the volume inside the cylinder under the plane $z = 2$.
 - Find the volume above the plane, inside the cylinder, and below the plane $z = 2$.
57. [T] The following figure shows the region D bounded by the curves $y = \sin x$, $x = 0$, and $y = x^4$. Use a graphing calculator or CAS to find the x -coordinates of the intersection points of the curves and to determine the area of the region D . Round your answers to six decimal places.
[Figure: A region is bounded by $y = \sin x$, $y = x^4$, and $x = 0$.]
58. [T] The region D bounded by the curves $y = \cos x$, $x = 0$, and $y = x^3$ is shown in the following figure. Use a graphing calculator or CAS to find the x -coordinates of the intersection points of the curves and to determine the area of the region D . Round your answers to six decimal

places.

[Figure: A region is bounded by $y = \cos x$, $y = x^3$, and $x = 0$.]

59. Suppose that (X, Y) is the outcome of an experiment that must occur in a particular region S in the xy -plane. In this context, the region S is called the sample space of the experiment, and X and Y are random variables. If D is a region included in S , then the probability of (X, Y) being in D is defined as $P[(X, Y) \in D] = \iint_D p(x, y) dx dy$, where $p(x, y)$ is the joint probability density of the experiment. Here, $p(x, y)$ is a nonnegative function for which $\iint_S p(x, y) dx dy = 1$. Assume that a point (X, Y) is chosen arbitrarily in the square $[0, 3] \times [0, 3]$ with probability density $p(x, y) = \begin{cases} \frac{1}{9}, & (x, y) \in [0, 3] \times [0, 3], \\ 0, & \text{otherwise.} \end{cases}$ Find the probability that the point (X, Y) is inside the unit square and interpret the result.
60. Consider X and Y , two random variables with probability densities $p_1(x)$ and $p_2(y)$, respectively. The random variables X and Y are said to be independent if their joint density function is given by $p(x, y) = p_1(x)p_2(y)$. At a drive-thru restaurant, customers spend, on average, 3 minutes placing their orders and an additional 5 minutes paying for and picking up their meals. Assume that placing the order and paying for/picking up the meal are two independent events X and Y . If the waiting times are modeled by the exponential probability densities $p_1(x) = \begin{cases} \frac{1}{3}e^{-x/3}, & x \geq 0, \\ 0, & \text{otherwise,} \end{cases}$ and $p_2(y) = \begin{cases} \frac{1}{5}e^{-y/5}, & y \geq 0, \\ 0, & \text{otherwise,} \end{cases}$ respectively, the probability that a customer will spend less than 6 minutes in the drive-thru line is given by $P[X + Y \leq 6] = \iint_D p(x, y) dx dy$, where $D = \{(x, y) \mid x \geq 0, y \geq 0, x + y \leq 6\}$. Find $P[X + Y \leq 6]$ and interpret the result.
61. [T] The Reuleaux triangle consists of an equilateral triangle and three regions, each of them bounded by a side of the triangle and an arc of a circle of radius s centered at the opposite vertex of the triangle. Show that the area of the Reuleaux triangle in the following figure of side length s is $\frac{s^2}{2}(\pi - \sqrt{3})$.
[Figure: An equilateral triangle with additional regions consisting of three arcs of a circle with radius equal to the length of the side of the triangle. These arcs connect two adjacent vertices, and the radius is taken from the opposite vertex.]
62. [T] Show that the area of the lunes of Alhazen, the two blue lunes in the following figure, is the same as the area of the right triangle ABC . The outer boundaries of the lunes are semicircles of diameters AB and BC , respectively, and the inner boundaries are formed by the circumcircle of the triangle ABC .
[Figure: A right triangle with points A , B , and C . Point B has the right angle. There are two lunes drawn from A to B and from B to C with outer diameters AB and BC , respectively, and with the inner boundaries formed by the circumcircle of the triangle ABC .]